

SP ENGINEERING ACADEMY

Video Lectures / Course Materials

COMPLETE COURSE

DIGITAL SYSTEMS

DIGITAL CIRCUIT DESIGN

Part – V – Standard Representation for Logic Functions

SOP Form and POS Form

Topic #	Content	
48	Standard Representation for Logic Functions - SOP Form and POS Form - https://youtu.be/RFC5LZstN5M	02
49	Standard SOP Form - Solved Problems - https://youtu.be/1U0xr9UOGeA	03
50	Standard POS Form - Solved Problems - https://youtu.be/FiBs2hMiCFc	07
51	M Notations - Minterms and Maxterms - https://youtu.be/GC2vo0BXbng	09
52	M Notations - Minterms and Maxterms - Solved Problems #01 - https://youtu.be/ig6mt4UBZPU	10
53	M Notations - Minterms and Maxterms - Solved Problems #02 - https://youtu.be/qaLorKcCam4	11

7. Standard Representation for Logic Functions

Boolean expressions are constructed by connecting the boolean constants and variables with the boolean operations. These Boolean expressions are also known as Boolean Formulae.

Video Link - <https://youtu.be/RFC5LZstN5M>

(i) Sum of Product Form (SOP)

The word Sum and product are derived from the symbolic representations of the OR and AND functions by + or . .

$$\text{a) } f(A, B, C) = \overline{A}B\overline{C} + A\overline{B}\overline{C}$$

↑ ↑
Product terms

$$\text{b) } f(P, Q, R, S) = \overline{P}Q + QR + RS$$

↑ ↑ ↑
Product terms

Also known as disjunctive normal form or disjunctive normal formula

(ii) Product of Sum Form (POS)

A product of Sums is any groups of Sum ANDed together.

$$\text{a) } f(A, B, C) = (A+B) \cdot (\overline{B}+C)$$

↑ ↑
Sum terms

$$\text{b) } f(P, Q, R, S) = (P+Q) \cdot (R+\overline{S}) \cdot (P+S)$$

↑ ↑ ↑
Sum terms

Also known as Conjunctive normal form or Conjunctive normal formulae

(iii) Standard SOP and Standard POS Forms

We can realize that in the SOP form, all the individual terms do not involve all literals.

Video Link - <https://youtu.be/1U0xr9UOGeA>

For example, in expression $AB + AB\bar{C}$ the first product term do not contain literal C .

If each term in SOP form contains all the literals then the SOP form is known as Standard or Canonical SOP form

Standard sum of product expression shown below

$$f(A, B, C) = \bar{A}\bar{B}C + AB\bar{C} + \bar{A}B\bar{C}$$

$\uparrow \quad \quad \uparrow \quad \quad \uparrow$
└──────────┘

Each product term consists of all literals in either complemented form or uncomplemented form

Each individual term in the Standard SOP form is called minterm

Product of Sum

Standard ~~sum of product~~ expression is shown below

$$f(A, B, C) = (A+B+C) \cdot (A+\bar{B}+C)$$

$\uparrow \quad \quad \uparrow$
└──────────┘

Each sum term consists of all literals in either complemented form or uncomplemented form

If each term in POS form contains all the literals then the POS form is known as Standard or Canonical POS form

Each individual term in the Standard POS form is called maxterm

(iv) Converting Expressions in Standard SOP or POS form

(a) Steps to Convert SOP to Standard SOP Form

- Find the missing literal in each product-term if any
- AND each product term having missing literal/s with term/s form by ORing the literal and its complements
- Expand the terms by applying distributive law and reorder the literals in the product-terms
- Reduce the expression by omitting repeated product-terms if any

(b) Steps to Convert POS to Standard POS form

- Find the missing literal in each sum term if any
- OR each sum term having missing literal/s with term/s form by ANDing the literal and its complements
- Expand the terms by applying distributive law and reorder the literals in the sum terms
- Reduce the expression by omitting repeated sum terms if any

Page _____

Example #01

Convert the given expression in Standard SOP form

$$f(A, B, C) = AC + AB + BC$$

→ Find the missing literal/s in each product-term

$$f(A, B, C) = AC + AB + BC$$

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└─┬─┬─ Literal A is missing
  └─┬─ Literal B is missing
    └─ Literal C is missing

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Original Product- článek

$f(A, B, C) = AC \cdot (B + \bar{B}) + AB \cdot (C + \bar{C}) + BC \cdot (A + \bar{A})$

Missing literals and their components

↑ AND product-term with missing literals & components

→ Expand the terms and reorder literals

$$f(A, B, C) = ABC + A\bar{B}C + AB\bar{C} + A\bar{B}\bar{C} + \bar{A}BC + \bar{A}B\bar{C}$$

$$f(A, B, C) = ABC + \bar{A}BC + A\bar{B}C + AB\bar{C}$$

Example #02

Convert the given expression in Standard SOP Form.

$$f(A, B, C) = A + ABC$$

$$\rightarrow f(A, B, C) = A + ABC$$

$\hookrightarrow$ Literals B and C are missing

$$f(A, B, C) = A \cdot (B + \bar{B}) \cdot (C + \bar{C}) + ABC$$

$$= ABC + AB\bar{C} + A\bar{B}C + A\bar{B}\bar{C} + ABC$$

$$= \underline{ABC + AB\bar{C} + A\bar{B}C + A\bar{B}\bar{C}}$$

Example #03

Express the following function in Standard SOP form

$$f(A, B, C, D) = AB + \bar{C}D + A\bar{B}C$$

$$f(A, B, C, D) = AB(C + \bar{C}) \cdot (D + \bar{D}) + \bar{C}D(A + \bar{A}) \cdot (B + \bar{B}) + A\bar{B}C(D + \bar{D})$$

$$\begin{aligned} &= ABCD + ABC\bar{D} + AB\bar{C}D + AB\bar{C}\bar{D} \\ &\quad + \bar{C}DAB + \bar{C}D\bar{A}B + \bar{C}DA\bar{B} + \bar{C}D\bar{A}\bar{B} \\ &\quad + A\bar{B}CD + A\bar{B}C\bar{D} \end{aligned}$$

Example #04

Determine the Canonical form of

$$f(x, y, z) = (xy + \bar{z})(y + x\bar{z})$$

$$f(x, y, z) = (xy + \bar{z})(y + x\bar{z})$$

$$= xy y + xy x \bar{z} + \bar{z} y + \bar{z} x \bar{z}$$

$$= xy + xy\bar{z} + y\bar{z} + x\bar{z}$$

$$= xy \cdot (z + \bar{z}) + xy\bar{z} + y\bar{z}(x + \bar{x}) + x\bar{z}(y + \bar{y})$$

$$f(x, y, z) = xyz + xy\bar{z} + x\bar{y}z + x\bar{y}\bar{z} + \bar{x}yz + \bar{x}y\bar{z}$$

$$= \underline{xyz + xy\bar{z} + \bar{x}yz + \bar{x}y\bar{z}}$$

Standard POS form

Video Link - <https://youtu.be/FiBs2hMiCFc>

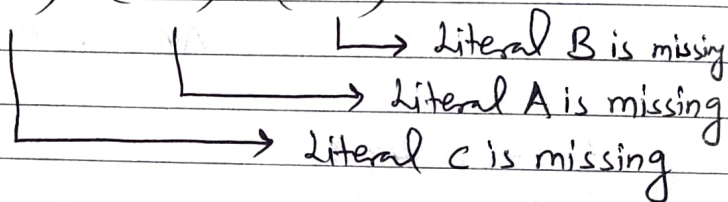
Example #01

Convert the given expression in Standard POS form

$$f(A, B, C) = (A+B) \cdot (B+C) \cdot (A+C)$$

→ Find the missing literal/s in each sum term

$$f(A, B, C) = (A+B) \cdot (B+C) \cdot (A+C)$$



→ OR Sum term with missing literals is its complement

$$f(A, B, C) = (A+B+C)(A+B+\bar{C})(B+C+A)(B+C+\bar{A})(A+C+B)(A+C+\bar{B})$$

$$\text{Since } A+B+C = (A+B)(A+C)$$

$$\therefore f(A, B, C) = (A+B+C)(A+B+\bar{C})(B+C+A)(B+C+\bar{A})(A+C+B)(A+C+\bar{B})$$

$$= (A+B+C)(A+B+\bar{C})(\bar{A}+B+C)(A+\bar{B}+C)$$

Example #02

Convert the given expression in Standard POS form

$$Y = A \cdot (A+B+C)$$

$$f(A, B, C) = A \cdot (A+B+C)$$

→ Literals B and C are missing

$$f(A, B, c) = (A + B \cdot \bar{B} + c \cdot \bar{c}) (A + B + c)$$

Since $A + Bc = (A + B)(A + c)$

$$\begin{aligned} \therefore f(A, B, c) &= (A + B\bar{B} + c) (A + B\bar{B} + \bar{c}) (A + B + c) \\ &= (A + B + c) (A + B + c) (A + \bar{B} + c) \\ &\quad (A + B + \bar{c}) (A + \bar{B} + \bar{c}) \\ &= (A + B + c) (A + \bar{B} + c) (A + B + \bar{c}) \\ &\quad (A + \bar{B} + \bar{c}) \end{aligned}$$

Example #03

Convert the given expression in standard POS form

$$f(P, Q, R) = (P + \bar{Q})(P + R)$$

$$f(P, Q, R) = (P + \bar{Q})(P + R)$$

$\swarrow$ Literal Q is missing
 $\searrow$ Literal R is missing

$$\begin{aligned} \therefore f(P, Q, R) &= (P + \bar{Q}) + (R + \bar{R}) (P + R) + (Q + \bar{Q}) \\ &= (P + \bar{Q} + R)(P + \bar{Q} + \bar{R})(P + R + Q)(P + R + \bar{Q}) \\ &= \underline{(P + Q + R)(P + \bar{Q} + R)(P + \bar{Q} + \bar{R})} \end{aligned}$$

M Notations - Minterms and Maxterms

Each individual term in Standard SOP form is called minimum minterm and each individual term in Standard POS form is maxterm

Video Link - <https://youtu.be/GC2vo0BXbng>

The Concept of minterms and maxterms allows us to introduce a very convenient shorthand notations to express logical functions

Minterms and maxterms for three Variables

Variables A B C	Minterms m_i	Maxterms M_i
0 0 0	$\bar{A}\bar{B}\bar{C} = m_0$	$A + B + C = M_0$
0 0 1	$\bar{A}\bar{B}C = m_1$	$A + B + \bar{C} = M_4$
0 1 0	$\bar{A}B\bar{C} = m_2$	$A + \bar{B} + C = M_2$
0 1 1	$\bar{A}BC = m_3$	$A + \bar{B} + \bar{C} = M_3$
1 0 0	$A\bar{B}\bar{C} = m_4$	$\bar{A} + B + C = M_4$
1 0 1	$A\bar{B}C = m_5$	$\bar{A} + B + \bar{C} = M_5$
1 1 0	$AB\bar{C} = m_6$	$\bar{A} + \bar{B} + C = M_6$
1 1 1	$ABC = m_7$	$\bar{A} + \bar{B} + \bar{C} = M_7$

$$f(A, B, C) = \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}\bar{C}$$

$$= m_0 + m_1 + m_2 + m_4$$

$$= \sum m(0, 1, 2, 4)$$

$$f(A, B, C) = (A + B + \bar{C})(A + \bar{B} + \bar{C})(\bar{A} + B + C)$$

$$= M_1 \cdot M_3 \cdot M_6 = \prod M(1, 3, 6)$$

$\sum$ - Sum of Product $\prod$ - Product of Sum

Example #01

Video Link - <https://youtu.be/liq6mt4UBZPU>

Express $F = A + B'C$ as sum of minterms

$$\begin{aligned} A + \bar{B}C &= A(B + \bar{B})(C + \bar{C}) + (A + \bar{A})\bar{B}C \\ &= (AB + A\bar{B})(C + \bar{C}) + A\bar{B}C + \bar{A}\bar{B}C \\ &= ABC + AB\bar{C} + A\bar{B}C + A\bar{B}\bar{C} + A\bar{B}C + \bar{A}\bar{B}C \\ &= ABC + AB\bar{C} + A\bar{B}C + A\bar{B}\bar{C} + \bar{A}\bar{B}C \\ &= m_7 + m_6 + m_5 + m_4 + m_1 \\ &= \sum m(1, 4, 5, 6, 7) \end{aligned}$$

Example #02

Express the boolean function $F = XY + \bar{X}Z$ in product of maxterms

$$\begin{aligned} F = XY + \bar{X}Z &= XY(Z + \bar{Z}) + \bar{X}Z(Y + \bar{Y}) \\ &= XYZ + XY\bar{Z} + \bar{X}YZ + \bar{X}\bar{Y}Z \\ &= m_7 + m_6 + m_3 + m_1 \\ &= \sum m(1, 3, 6, 7) \\ &= \prod M(0, 2, 4, 5) \end{aligned}$$

Example #03

Express the Boolean function as (i) POS form (ii) SOP form

$$D = (A' + B)(B' + C)$$

(i) POS form $D = (\bar{A} + B)(\bar{B} + C)$

(ii) SOP form $D = (\bar{A} + B)(\bar{B} + C)$

$$= \bar{A}\bar{B} + \bar{A}C + B\bar{B} + BC$$

$$= \bar{A}\bar{B} + \bar{A}C + BC \quad [\because B\bar{B} = 0]$$

Example # 04Video Link - <https://youtu.be/qaLorKcCam4>

Express the following function in product of maxterms
 $F(x, y, z) = (xy + z)(y + xz)$

$$F(x, y, z) = (xy + z)(y + xz)$$

$$= xy + xz + y + yz$$

$$= xy + xz + y + yz$$

$$= xy(1 + z) + y + yz$$

$$= xy + y + yz$$

$$= xy(z + \bar{z}) + y(z + \bar{z}) + xz(y + \bar{y})$$

$$= xyz + xy\bar{z} + xz + \bar{x}yz + x\bar{y}z + x\bar{y}\bar{z}$$

$$= xyz + xy\bar{z} + x\bar{y}z + \bar{x}yz$$

$$= m_7 + m_6 + m_5 + m_3$$

$$F(x, y, z) = \sum m(3, 5, 6, 7)$$

$$F(x, y, z) = \prod M(0, 1, 2, 4)$$

$$F(x, y, z) = (x + y + z)(x + y + \bar{z})(x + \bar{y} + z)(\bar{x} + y + z)$$

Example #05

Obtain the truth table of the function

$$F = xy + x\bar{y} + \bar{y}z$$

$$F = xy + x\bar{y} + \bar{y}z = xy(z + \bar{z}) + x\bar{y}(z + \bar{z}) + \bar{y}z(x + \bar{x})$$

$$= xyz + xy\bar{z} + x\bar{y}z + x\bar{y}\bar{z} + x\bar{y}z + \bar{x}\bar{y}z$$

$$= xyz + xy\bar{z} + x\bar{y}z + x\bar{y}\bar{z} + \bar{x}\bar{y}z$$

x	y	z	xyz	xy $\bar{z}$	x $\bar{y}$ z	x $\bar{y}\bar{z}$	$\bar{x}\bar{y}$ z	output
0	0	0	0	0	0	0	0	0
0	0	1	0	0	0	0	1	1
0	1	0	0	0	0	0	0	0
0	1	1	0	0	0	0	0	0
1	0	0	0	0	1	1	0	1
1	0	1	0	0	1	0	0	1
1	1	0	0	1	0	0	0	1
1	1	1	1	0	0	0	0	1